



Carla Ferreira

Cardiac Physiologist | Biomedical Engineer | Data Scientist

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PROFESSIONAL SUMMARY

Cardiac Physiologist with an MSc in Biomedical Engineering and a completed Postgraduate Program in Enterprise Data Science and Analytics. Experienced in translating complex clinical and business datasets into actionable insights through statistical analysis, predictive modeling and interactive dashboards. Combines healthcare domain expertise with technical skills in Python, SQL and cloud platforms. Seeking a junior data role to apply analytical capabilities and contribute to data-driven decision-making in a collaborative environment.

TECHNICAL SKILLS

- Programming & Analytics: Python (Pandas, Scikit-learn, XGBoost, Statsmodels), R, SQL (T-SQL, PostgreSQL), DAX
- Cloud & Big Data: AWS (S3, Glue, Athena), Azure (Cosmos DB, Databricks), Microsoft Fabric, DuckDB
- Databases: SQL Server, NoSQL (MongoDB, Cassandra, Table API), Query Optimization, Dimensional Modeling
- Machine Learning: Predictive Modeling, Feature Engineering, SHAP/LIME, Hyperparameter Tuning, SMOTE
- Visualization & BI: Power BI, Dash, Plotly, Matplotlib, Seaborn, Interactive Dashboards
- Tools & Development: Git, Jupyter, Anaconda, Hugging Face, VS Code, Polars

EDUCATION

Postgraduate Program in Enterprise Data Science and Analytics | Feb 2025 - Feb 2026

NOVA Information Management School (NOVA IMS), Universidade Nova de Lisboa

Final grade: 18

Master's Degree in Biomedical Engineering | Sept 2020 - May 2024

Instituto Superior de Engenharia de Lisboa (ISEL), Instituto Politécnico de Lisboa

Final grade: 18

Bachelor's Degree in Clinical Physiology | Sept 2015 - Sept 2019

Escola Superior de Tecnologia da Saúde de Lisboa (ESTeSL), Instituto Politécnico de Lisboa

Final grade: 14

PROFESSIONAL EXPERIENCE

Cardiac Physiologist

Serviços Sociais da Câmara Municipal de Lisboa | Feb 2025 - Present

- ECG monitoring
- Holter final report
- Performing Stress Test

Hospital de Santa Cruz, Centro Hospitalar de Lisboa Ocidental | June 2020 - Present

- ECG, ABPM and Tilt Test monitoring
- Holter monitoring and final report
- Performing Stress Test, Cardiorespiratory Stress Test, Exercise Stress Test and/or Pharmacologic Stress Test in the context of Myocardial Scintigraphy
- Follow-up consultation and remote monitoring assessment for patients with implantable cardiac devices
- Streamlined operational workflows by developing data management tools for scheduling, inventory tracking and resource allocation, improving team efficiency and reducing administrative overhead

- ECG, Holter and ABPM monitoring
- Performing Stress Test, Polysomnography and Pulmonary Function Tests

MOST RELEVANT ACADEMIC DATA SCIENCE PROJECTS

Retail Sales Analytics & Business Intelligence | Power BI, Microsoft Fabric, DAX

- Developed an end-to-end business intelligence solution for multi-location retail data, designing a star schema semantic model and implementing multiple DAX measures for sales and profitability analysis.
- Built interactive Power BI dashboards to monitor revenue trends, profit margins and product performance, supporting data-driven decisions in inventory management and pricing strategy.

Employee Attrition Risk Modeling | Python, XGBoost, SMOTE, Hugging Face

- Built and deployed XGBoost classification model for employee turnover prediction on Hugging Face platform, enabling HR teams to identify at-risk employees
- Generated actionable insights using SHAP/LIME explainability techniques, revealing overtime and job satisfaction as primary attrition drivers

Colorectal Cancer Survival Prediction | Python, Scikit-learn, SHAP, LIME

- Evaluated and compared multiple classification models on a real-world clinical dataset, improving performance through feature engineering and hyperparameter tuning.
- Applied explainable AI techniques (SHAP and LIME) to identify key predictive factors such as tumor size, cancer stage and treatment type, supporting model interpretability in a clinical context.

Database Design & Auction System | SQL, T-SQL, Azure

- Architected auction system with stored procedures, transaction handling and automated bid validation for concurrent users.
- Conducted market analysis to identify optimal store locations using sales patterns and demographic data.

Big Data Analysis & Cloud Infrastructure | AWS (S3, Glue, Athena), DuckDB, Polars

- Processed 32M movie ratings using cloud pipeline (AWS) and local engines (DuckDB/Polars) from 100K to 32M records.
- Optimized query performance through Parquet format and partitioning, reducing processing time for large aggregations.

CERTIFICATIONS

- MEB Diploma of Merit - Ordem dos Engenheiros, 2025
- Good Clinical Practice Certification for Alpine 1 Clinical Trial participation - Hospital de Santa Cruz, 2025
- Speaker, VI Jornadas de Cardiopneumologia, Workshop on Cardiopulmonary Exercise Testing - Hospital Prof. Dr. Fernando Fonseca, 2025
- Quantitative Research Methodology in Health Sciences - Hospital de Santa Cruz, 2023

DEVICE RELATED TRAINING

- Implantable Cardioverter/Defibrillator Course - Medtronic, Oct 2025
- CareLink Course - Medtronic, 2024
- Pacemaker Course - Medtronic, 2024
- Electrocardiography in Cardiac Pacing - Medtronic, 2024
- Conduction System Pacing Course Medtronic, 2024
- Micra-AV Follow-up (Medtronic) - 2023
- Introduction to Pacemaker Follow-up - Medtronic, 2021

LANGUAGES

Portuguese (Native)	English (Proficient)	Spanish (Beginner)	French (Beginner)
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